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Use of superpixels for improvement of inter-rater and intra-rater reliability during annotation of medical images

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Highlights

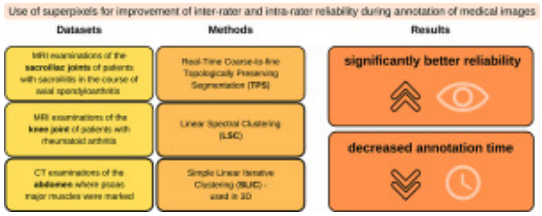
- Superpixel presegmentation is proposed for preparing segmentation ground truth data.
- Superpixel-based annotations are compared to manual ones on multiple criteria.
- Applied method allows to significantly reduce annotation time (up to 2x).
- Application of superpixels significantly increases inter- and intra-rater agreement.

Abstract

In the context of automatic medical image segmentation based on statistical learning, raters' variability of ground truth segmentations in training datasets is a widely recognized issue. Indeed, the reference information is provided by experts but bias due to their knowledge may affect the quality of the ground truth data, thus hindering creation of robust and reliable datasets employed in segmentation, classification or detection tasks. In such a framework, automatic

medical image segmentation would significantly benefit from utilizing some form of presegmentation during training data preparation process, which could lower the impact of experts' knowledge and reduce time-consuming labeling efforts. The present manuscript proposes a superpixels-driven procedure for annotating medical images. Three different superpixeling methods with two different number of superpixels were evaluated on three different medical segmentation tasks and compared with manual annotations. Within the superpixels-based annotation procedure medical experts interactively select superpixels of interest, apply manual corrections, when necessary, and then the accuracy of the annotations, the time needed to prepare them, and the number of manual corrections are assessed. In this study, it is proven that the proposed procedure reduces inter- and intra-rater variability leading to more reliable annotations datasets which, in turn, may be beneficial for the development of more robust classification or segmentation models. In addition, the proposed approach reduces time needed to prepare the annotations.

Graphical abstract



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Keywords

Inter-rater reliability; Superpixel segmentation; Image annotation; Data quality; Diagnostic imaging; Deep learning training

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1. Introduction

Medical image analysis is a major interest in biomedical research (Ayache and Duncan, 2016). Among the several topics tackled, one can

mention segmentation, visualization, quantification, object tracking, and detection (Chen et al., 2018), which are strictly linked with the applications in the area of medical image diagnosis.

Evaluation and validation of new methods for medical image analysis have been based for a long time on personal datasets, so that unbiased and reliable comparison of the new solutions to the existing ones was hard to perform (Price, 1986).

Nowadays, international challenges have become the standard for validation of medical image analysis methods. However, the lack of quality control on such challenges may have substantial consequences in terms of reproducibility and interpretation of the results (Maier-Hein et al., 2018). Also, the large variability in challenge design may affect the reliability of the validation. Indeed, several issues have been noticed (Maier-Hein et al., 2018) such as (i) minor changes in metrics may make the last the first; (ii) different data aggregation methods produce different winners; (iii) different annotators produce different winners; (iv) removing one test case can change the winner.

In such a context, the need for better ways to collect, annotate, and reuse medical imaging data is clear. The first key point for such a problem is that the annotation of each image must be as accurate and reproducible as possible (Kohli et al., 2017).

In fact, supervised ground-truth images are provided by experts but present several issues, in terms of time of the annotation procedure, raters reliability, and bias due to experts' knowledge. Rater-related variability of annotations is a well recognized issue among researchers. To address this problem to some extent, the tendency is to find ways to efficiently combine different annotations (Renard et al., 2020). Converting usually small number of annotations into ground truth data affects the robustness of trained models and limits the possibility to apply the same approaches to different situations/datasets. Conversely, raters' agreement issues could be addressed by enlarging the pool of annotators, however, it may result in a hard and time-consuming task.

The combination of multiple annotations into a single fused ground truth need not to be beneficial for subsequent tasks e.g. segmentation. Indeed, it has been shown (Lemay et al., 2022) that deep segmentation models trained on multiple annotations of the same structure estimate better the segmentation uncertainty than segmentation models trained on fused annotations. For this reason it

is reasonable to search for annotation methods which decrease inter- and intra-reader annotation variability as this will in turn reduce uncertainty of segmentation models trained on such a data.

In such a context, it may be expected that supervised medical image segmentation tasks would benefit from standardizing the process of ground-truth preparation supported by e.g. some form of pre-segmentation (not necessarily problem-specific), resulting in reducing the intra- and inter-reader variability, decreasing the time needed to annotate the data, and allowing the expert to focus on peculiar areas where to add their knowledge.

The present manuscript proposes a superpixels-driven procedure for annotating medical images, consisting of a pre-processing phase based on superpixels. Superpixels are groups of pixels sharing some characteristics, without semantic meaning necessarily associated. Their use in the contexts of Image Processing and Computer Vision is significant, as they enable a comprehensive understanding of local properties of the image, without basing on punctual information.

In the proposed annotation pipeline, medical experts are required to select the superpixels of interest in an interactive fashion, and then to manually correct the annotation if – in the opinion of the rater – it does not coincide with the borders of superpixels. Specifically, the original data is over-segmented via a superpixel algorithm, producing a grid to be superimposed on the annotated image. An underlining labels image is produced, to enable superpixels annotations. The final output consists of the binary annotation by the expert. Such a procedure is supposed to reduce intra- and inter-rater variability and the dependence on the skill of the annotators, thus enabling more robust classifiers, trained on such labeled data. In addition, it should reduce time needed for manual annotation. The presented study was performed by the development and use of an annotation tool. Indeed, a tool facilitating the annotation allows to widen the pool of annotators, enabling the creation of datasets with more different labels and therefore more robust and reliable. As a further future development of such an approach, a web service providing the aforementioned tool would enhance the accessibility by raters, thus granting a large number of annotations.

The use of superpixels and the definition of a procedure for constructing reliable datasets can also foster the development of Deep Learning based classifiers or detector, to be deployed for image diagnosis purposes. In the literature, superpixels and deep learning

can be jointly found in (i) methodological frameworks (Lee et al., 2017, Zhang et al., 2018, Yang et al., 2020, Suzuki, 2021); (ii) Remote Sensing; and (iii) Biomedical Imaging. They are deployed in different phases and for several purposes, e.g., as post processing for the refinement of the output of the neural networks (Cao et al., 2016, Kawahara and Hamarneh, 2018); as pre-processing for working on patches (Wang et al., 2017) and guiding the feature extraction process (Fan et al., 2017); to have correct delineation of objects and perform the classification in a dense way but with good computing times (Postadjian et al., 2018); to completely avoid annotations (self-supervised algorithms) (Ouyang et al., 2020); to improve classification based on manual annotations (Li et al., 2020, Wang et al., 2019).

Despite the relevant interest of the scientific community towards the presented topic, to the best of the authors' knowledge this is the first tentative to leverage on superpixels for facilitating the annotation process and improving raters' agreement, in view of the deployment of deep learning techniques.

Among the methods for superpixels segmentation, three have been considered in the present work. They are described in Section 2.2 with regards to their choice for being the supporting algorithm in the annotation process. Details of these superpixel algorithms can be found in the corresponding papers (Chen et al., 2017, Yao et al., 2015, Achanta et al., 2010, Achanta et al., 2012).

The proposed study is applied to three datasets: two different MRI datasets, regarding STIR images of subchondral bones, aimed at highlighting spondyloarthritis lesions, contrasted T1-weighted images of the knee, aimed at detecting synovitis, and CT abdominal images, aimed at detecting psoas major muscles.

The major novelties of the proposed work are:

- The evaluation of using superpixels for supporting manual annotations of objects of interest in medical images.
- The assessment of inter-rater reliability.
- The estimation of the time needed for the annotation process.
- The estimation of the degree of manual corrections to superpixels.

- The assessment of intra-rater reliability.
- The application to three different clinical tasks.

The paper is structured as follows. Section 2 is for the description of the study: in particular, the datasets, the used methods, the labeling procedure, and the evaluation metrics are introduced. Section 3 describes the experimental phase for studying the effects of the superpixel-driven annotation on the labeling procedure. Results are reported in Section 4. Section 5 is for discussion and conclusion of the work.

2. Materials and methods

2.1. Materials

Overall, 15 3D magnetic resonance imaging (MRI) examinations of the joints were included in this retrospective study – 9 3D MRI examinations of the sacroiliac joints of patients with sacroiliitis in the course of axial spondyloarthritis (181 2D cross-sections) and 6 3D MRI examinations of the knee joint of patients with rheumatoid arthritis (in total 205 2D cross-sections, but the annotations were performed only on 48 cross-sections where epiphyses of the femur and tibia were clearly visible). Moreover, two additional training cases (one case for each anatomic location) were used to unify annotation procedure among different raters, as described in detail in the following chapter. All examinations were acquired in 3.0 T MRI scanner (Achieva, Philips Healthcare, Amsterdam, Netherlands).

In the case of MRI of the sacroiliac joints (SIJ), only STIR (short tau inversion recovery) sequence was further analyzed. Both sacroiliac joints were imaged simultaneously from the anterior to the posterior border in the coronal oblique plane, parallel to the long axis of the sacrum with the use of 8-channel phased-array XL-torso body matrix coil. The detailed protocol of STIR turbo spin echo sequence was: TR (repetition time) 5239 ms, TE (echo time) 30 ms, inversion time 190 ms, flip angle 90°, NEX (number of excitations) 2, slice thickness 3 mm, matrix 400 × 400, FOV (field of view) 240 × 240

The analysis of MRI examination of the knee was performed on subtraction images from pre- and post-contrast T1-weighted sequences acquired in the coronal plane with the use of 8-channel knee coil. The detailed protocol of pre-contrast T1-weighted turbo

spin echo sequence was: TR 500–700 ms, TE 20 ms, flip angle 90°, NEX 1, slice thickness 3 mm, matrix 240 × 214, FOV 180 × 150. In order to obtain post-contrast T1-weighted sequence, intravenous contrast agent Gadobutrolum (Gadovist, Bayer, Germany) was administered. The protocol of post-contrast T1-weighted turbo spin echo sequence was: TR 710 ms, TE 20 ms, flip angle 90°, NEX 1, slice thickness 3 mm, matrix 360 × 291, FOV 180 × 150. Original pre- and post-contrast T1-weighted sequences of each patient were available to comparison in the window next to the analyzed subtraction image.

As an additional imaging modality, whole-body PET-CT examinations of Head and Neck Squamous Cell Carcinoma patients were also analyzed. They were retrieved from Head and Neck Cancer CT dataset available in "The Cancer Imaging Archive" (Grossberg et al., 2017). From 430 available 3D examinations, 6 were randomly chosen for further analysis. All scans were obtained using GE Medical Systems Discovery ST or RX hybrid PET/CT scanners after the intravenous injection of 18F-labeled fluorodeoxyglucose. Following parameters were used in examinations acquisition: voltage 120 kVp; current 300 mA; rotation time 0.5 s; slice thickness 3.75 mm; size 512 K; resolution 1.0240 pixels per mm; pixel size 0.9766 mm × 0.9766 mm. Every examination incorporated different number of 2D axial slices and only those located between upper part of pelvic bone and bottom part of the lung were labeled as volume of interest (VOI) and analyzed in our study. The VOI was further restricted to the part including psoas major muscles and finally 60 2D cross-sections were analyzed in this study.

In order to ensure reproducibility, custom datasets were made publicly available at: https://data.mendeley.com/datasets/425fcppvkn ↗

2.2. Superpixeling methods

In digital image processing, the traditional goal of segmentation is to partition the image into a finite number of regions that possess a certain homogeneity property based on the definition on the homogeneity predicate. The pixels in a region form a so-called connected component which will hopefully play a role in localizing the content of visual information of interest.

As a consequence, a good segmentation is one characterized by a good match between a segmented region and a region of interest (ROI). An oversegmentation error occurs when the predicate is true for the

union of two adjacent Regions; an undersegmentation error occurs when the predicate is false for the given region (Dellepiane, 1991). The errors of oversegmentation and undersegmentation usually appear at the same time and a compromise is usually sought.

The accuracy of a segmentation result can be evaluated by computing some measure of agreement with corresponding ground-truth data defined a-priori, when available, or it can be computed from some properties of the image. In general, good segmentation shows a good match between the edge of the segmented regions and the gradients of the image (boundary recall). Due to the general difficulties in achieving a good segmentation result when addressing the problem of delineating anatomical or pathological regions, some tools could be exploited interactively to facilitate the annotation process.

For instance, a pre-processing step might push the expert to annotate the regions of interest avoiding tracing outlines and speeding up the pixel selection and grouping phase, if the pre-processing result has good edge fidelity. In addition, the simplest interaction is based on manually clicking on the sub-portions of the region that needs to be outlined, given the algorithm have no undersegmentation errors. Oversegmentation errors are tolerated.

This is the case with the superpixel segmentation approach which greatly reduces undersegmentation errors while tolerating oversegmentation errors. A number of methods have recently been proposed in the pursuit of good performance related to reliable and consistent partitioning and processing speed. They are usually based on the optimization of some functions and may differ in the functional definition and optimization approach. The method “Real-Time Coarse-to-fine Topologically Preserving Segmentation” (Yao et al., 2015), hereinafter named TPS, was selected for its proven positive performance compared to other superpixel segmentation algorithms such as SLIC (Achanta et al., 2010) and SEEDS (Bergh et al., 2012) in 2015. It was also found to be the best superpixeling algorithm in benchmark performed in 2016 (Stutz et al., 2018). More recently (in 2017), in addition to SLIC and SEEDS, the method described in Chen et al. (2017), Linear Spectral Clustering (LSC), also outperformed other methods (e.g., Ncuts (Shi and Malik, 2000), ERS (Liu et al., 2011), and TurboPixel (Levinshtein et al., 2009)).

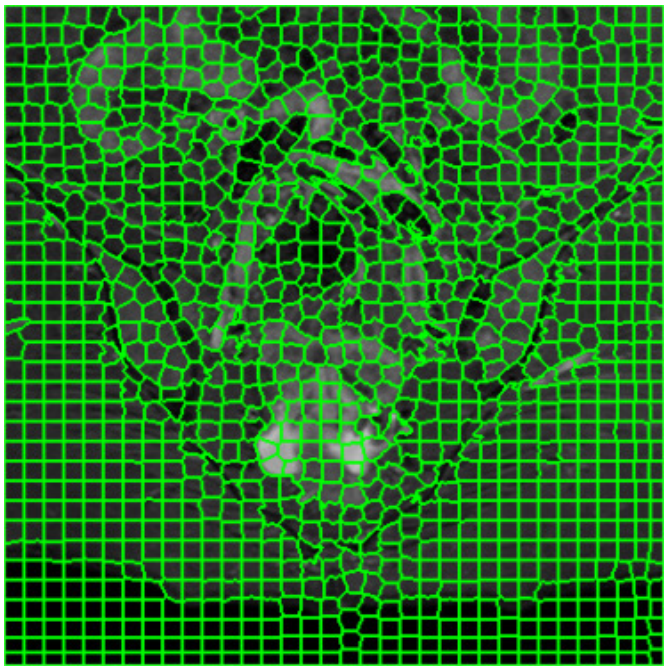
2.2.1. Real-Time Coarse-to-fine Topologically Preserving Segmentation method - TPS

The “Real-Time Coarse-to-fine Topologically Preserving Segmentation” (TPS) method defines the superpixel segmentation problem as a boundary and topology preserving Markov random field (MRF). The coarse-to-fine optimization technique accelerates the inference in terms of the number of updates by an order of magnitude. The objective function to be optimized here is similar to the k-means clustering function, but also incorporates morphological, topological, and dimensional information, to look for regular shape, connectivity, and small size. To avoid the counter-effect of tiny superpixels, size constraints are added. MRF energy is composed of 5 terms related to: regularity of the shape pushing the compactness; homogeneity of color; superpixel outline that encourages a small boundary length; topology forcing connection; minimum size.

The process relies on a coarse-to fine optimization scheme for the assignment task and a priority queue FIFO strategy that specifically processes boundary blocks to find instances where connectivity is breached.

The running parameters are: number of superpixels and coefficients defining weights for each compound of MRF energy, which allows to modify behavior of the algorithm. The same default weight was set for all energy components in our experiments, since it was believed that the only a-priori intervention should be driven by the medical experts performing the annotation (indeed, recall that the superpixel only provide a support to the manual annotation rather than the segmentation itself).

An exemplary division made using the method is shown in Fig. 1.



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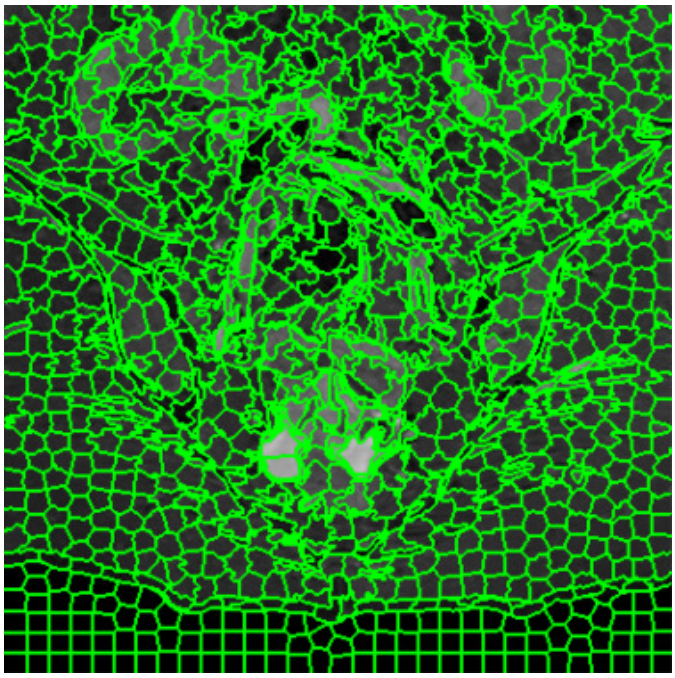
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Fig. 1. Sample division into superpixels using TPS method.

2.2.2. Linear Spectral Clustering - LSC

The Linear Spectral Clustering (LSC) algorithm (Chen et al., 2017) is a superpixel segmentation method granting high boundary adherence and visual compactness. The key point for its selection is that it preserves global image structures through efficient local operations. Also, form the implementation point of view, it possesses linear computational complexity and high memory efficiency, thus not demanding too much computational efforts for a pre-processing phase.

An exemplary division made using the method is shown in Fig. 2.

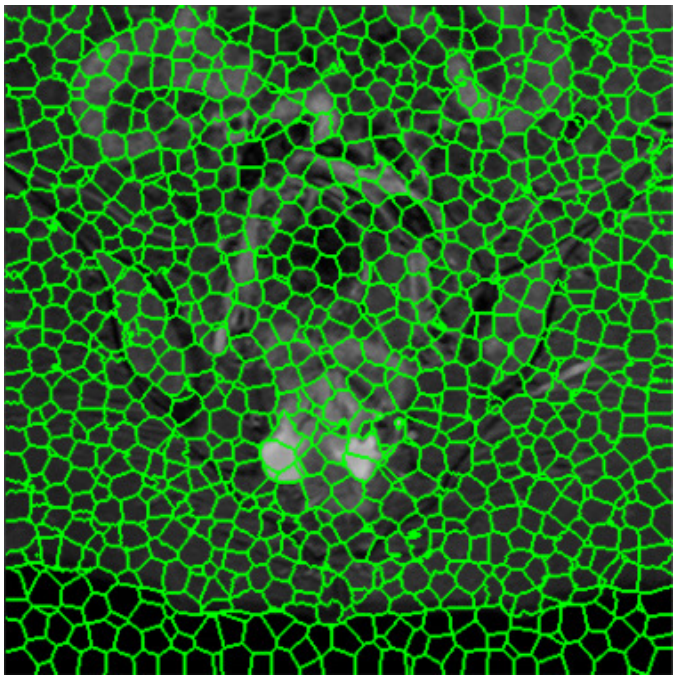


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Fig. 2. Sample division into superpixels using LSC method.

2.2.3. Simple Linear Iterative Clustering - SLIC 3D

To verify the performance of superpixeling methods operating in volume (segmentation into supervoxels) 3D version of popular SLIC algorithm (Achanta et al., 2010) was tested. For that purpose scikit-image implementation of the algorithm was used in SLIC-zero mode which does not require manual adjustment of algorithm parameters. SLIC superpixeling method is a fast and accurate algorithm for image analysis. It combines clustering and boundary adherence to produce visually coherent superpixels. SLIC initializes seed points on a regular grid, assigns pixels to the nearest seed point based on color and spatial proximity, and iteratively updates the seed points until convergence, refining the superpixel boundaries. It excels at boundary adherence, aligning superpixel boundaries with object boundaries. SLIC offers a balance between accuracy and computational efficiency, making it suitable for real-time applications. It has been successfully applied in various computer vision tasks, including object recognition, image segmentation, and video analysis (Achanta et al., 2012). An exemplary division made using the method is shown in Fig. 3.



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Fig. 3. Sample division into superpixels using SLIC 3D method (cross-section of generated supervoxels through image slice is visualized).

2.3. Annotations acquisition

The lesions were annotated by 4 doctors. In the case of the sacroiliac joints, the bone marrow edema was marked in MR images. In the MR images of the knee joints synovitis was indicated. In the case of the knee joints, the annotation area was limited to the epiphyses region only, which was determined for each slice by one of the physicians. The bounding box for this region was displayed at the time of the annotation. Finally, for CT abdominal data psoas major muscles were outlined.

Before starting the work on annotations a unification session was conducted using a single training case (not used in the remaining part of this study) for each of the two MR tasks. Each physician annotated lesions in these cases, and then the entire team of doctors analyzed the emerging differences and developed a common procedure and clear rules for what should be considered a lesion. Thanks to this, it was possible to exclude differences resulting from different opinions of individual doctors and to cleanly analyze the impact of the method used to mark the lesions. In the case of CT data such a training session was not necessary, as the definition of what constitutes a psoas major muscle is unequivocal. The annotations were performed in dedicated

software prepared for the purposes of this study. When the superpixels were used to draw annotations, doctors first marked the entire superpixel and then made manual corrections by freely adding/subtracting pixels from the originally selected area. When the SLIC 3D method was used, the supervoxel selected on one of the slices was marked on all slices through which it crossed. In addition, the time spent on annotating all slices for each patient was recorded.

In the first phase, for each of the cases, the annotations were made manually and with the use of superpixels generated by the TPS, LSC, and SLIC 3D methods. For each of the three methods, two superpixel images were created - with lower and higher number of superpixels - 1000/1250/1600 (hereinafter LSC-lower, TPS-lower and SLIC-lower), 2000/2500/3200 (referred further as LSC-higher, TPS-higher and SLIC-higher) for the sacroiliac joint, knee and abdomen, respectively (the discrepancies result from different image sizes - 400 × 400 px, 448 × 448 px and 512 × 512 px, respectively - and the purpose of their introduction was keeping a similar average number of pixels in a superpixel). In case of SLIC 3D method only the desired number of supervoxels is used as the algorithm input parameter. To make the comparison with other methods reliable, the number of supervoxels was chosen in such a manner that the average number of supervoxels intersecting the image slice was similar to the number of superpixels in other methods. Each annotation was repeated by 3 randomly chosen physicians, so each test case was reviewed a total of 21 times (7 methods $\times$ 3 annotations). That kind of approach was used to make the inference applicable to a wider population of raters (Gwet, 2014), not to just the three raters who actually annotated the case. Moreover in order to avoid the impact of repeated reviewing and “learning” the same test cases and the growing experience in working with new methods, an individual, fully randomized order of annotations was prepared for each physician.

In the second phase, the repeatability of the annotations performed by the same physician was checked. As during the first phase analysis the LSC method seemed to offer better outcome than the other two methods (even though some of the differences were statistically irrelevant), was faster and required less manual corrections, only this method was compared with the manual method for the purposes of the intra-rater reliability study (as in the first phase, annotations based on lower and higher number of superpixels were analyzed). Each of the physicians repeated their annotations three times – annotations were repeated for randomly selected three 3D cases of

the sacroiliac joints, two 3D cases of the knees and two 3D cases of the abdominal images for each method evaluated in this phase. This random selection of cases to be annotated was done separately for each physician.

2.4. Metrics description

As is known, the reliability analysis must be faced from different points of view. The inter- and intra-reader agreement between evaluators is a fundamental aspect to be tested when assessing an annotation method in order to prove the reliability of the procedure.

To this end, three similarity metrics are taken into account here, in order to demonstrate that the use of a proposed superpixel-based tool as a support for annotating medical data not only reduces the time required, but also promotes better agreement between evaluators (inter-rater reliability), and better replicability (intra-rater reliability).

The metrics used in the present study are two classical ones, i.e., the Dice's coefficient and the Cohen's Kappa coefficient (Chang et al., 2009), and a custom coefficient, hereinafter referred to as *Custom Index*, which is defined below.

The proposed Custom Index was designed based on agreement area ratios in order to have the possibility to take into consideration all the annotations at once. It is defined as:

$$CI = \frac{A\left(ann_1 \cap ann_2\right) + A\left(ann_1 \cap ann_3\right) + A\left(ann_2 \cap ann_3\right)}{A\left(ann_1\right) + A\left(ann_2\right) + A\left(ann_3\right)}, \quad (1)$$

with A being the number of pixels belonging to the set of pixels being the argument of A and ann_i denoting ith annotation of analyzed case. Recall that each case has been annotated by three raters.

In addition, for the sake of clarity, even though the Dice's coefficient and the Cohen's Kappa coefficient metrics are well-known, they are briefly introduced below.

Given a sample space Ω, let Y denote the set of Observed Positive elements and X the set of Predicted Positive elements. The Confusion Matrix is then used to take into account the True Positive (TP), the False Positive (FP), the False Negative (FN), and finally the True Negative (TN) subsets as:

- $TP = X \cap Y$;

- $FP = X \setminus Y$;
- $TN = \Omega \setminus \{X \cup Y\}$;
- $FN = Y \setminus X$.

By using such values, the Dice's coefficient is defined as:

$$DICE = \frac{2\,|TP|}{2\,|TP| + |FP| + |FN|}. \tag{2}$$

The Cohen's Kappa coefficient is defined as:

$$k = \frac{2\,(|TP\|TN| - |FP\|FN|)}{(|TP| + |FP|)\,(|FP| + |TN|) + (|TP| + |FN|)\,(|FN| + |TN|)}. \tag{3}$$

These two metrics are used here to evaluate the agreement between two raters: to this end X is associated with the positive elements of the first rater and Y with the positive elements of the second rater.

The Dice's coefficient is normalized to the range $[0, 1]$ and approaches 1 when there is strong agreement among the raters. Cohen's Kappa is in the interval $[-1, +1]$ and indicates a perfect agreement with 1, a bad agreement with lower values and a great disagreement with negative values.

It is worth mentioning that, since such indices refer to pairwise agreement, the following analysis will use their averages over all possible pairwise comparisons between couples of annotations, namely:

- Average pairwise Dice's coefficient: Dice's coefficient is computed for each possible couple of annotations for each annotated 2D image, and then the mean value of these Dice's coefficients are calculated for each case, i.e., $avg\,(DICE\,(ann_i, ann_j)) \quad i, j = 1, 2, 3 \text{and} i \neq j$ for each method, for each superpixel number.
- Average pairwise Cohen's K calculated in a similar fashion as average pairwise Dice's coefficient.

2.5. Applying method to a new task

The following steps should be carried out to analyze the usability of the proposed method on the new dataset:

1. Dataset Selection: Choose the new dataset that represents the specific task to which the method will be applied. Ensure the dataset includes the necessary images and lesions that require annotation.

2. Rater Selection: Select a team of raters, three or more physicians, who will perform annotations for the new dataset. This group of raters should represent a range of perspectives and expertise to ensure the method's applicability to a broader population of raters.

3. Unification Session: Conduct a unification session with a team of doctors. During this session, the doctors collaboratively annotate lesions in the new dataset, and any differences in interpretation are discussed and resolved. This step is crucial to establish clear and consistent identification rules for the new task.

4. Superpixels Preparation: Determine the superpixelization parameters for the new dataset (ensure these parameters align with the dataset's specific characteristics). Generate using all evaluated superpixeling methods.

5. Annotation Process: Have the chosen raters independently annotate the new dataset using the proposed method. Each image should be annotated at least 3 times, following the established annotation rules. The order of annotations should be randomly chosen for each rater. Time spend on each annotation should be noted.

6. Performance Evaluation: Calculate chosen quality metrics and run statistical analysis of the results (especially statistical significance tests). This evaluation includes comparing the agreement among raters, amount of manual corrections and the time cost.

7. Repeatability Analysis: In a separate phase, assess the repeatability of annotations made by the same rater. Each rater should annotate chosen images two more times (in randomized order) and intra-rater variability should be evaluated.

3. Experimental phase

As previously mentioned, the aim of the present study is to check how a superpixels-based tool supporting the annotation process facilitates the creation of robust and reliable datasets with labels. In particular, the primary goals of the planned experiments were to test the following hypotheses:

- the use of superpixels leads to more agreement between different raters (improves inter-rater agreement) compared to manual annotations;
- the use of superpixels leads to more agreement between the same rater when repeating the same annotation at different time slots (improves intra-rater agreement) compared to manual annotations.

The other goals of the experiments were:

- identifying which is the best superpixels-based method among the considered ones;
- identifying which is the best number of superpixels based on the considered method.

Therefore, the present study aims at showing the operational advantage in using the developed tool for the doctors, thus justifying the deployment of such an approach in further and wider studies.

3.1. Procedures

The operational procedure consists of two steps for each of the aforementioned goals:

- Compute the previously described similarity indices;
- Perform a statistical testing on those.

For the sake of clarity, the procedure is illustrated with the following example. Assume that the aim is to test whether the use of LSC with 1250 superpixels yields a better agreement between raters than the Manual approach. Because for the indices used in this study the higher the indices values, the more agreement between raters, the

null hypothesis $H0$ and the alternative hypothesis $H1$ are as follows:

$$H0 : Index_from_1250LSC = Index_from_Manual,$$

$$H1 : Index_from_1250LSC > Index_from_Manual.$$

Assume the focus on the knee dataset for which 48 2D images were collected for annotations. Then, four raters performed the annotations on different subsets of the knee set in such a way that three annotations are available for each of the above images for both Manual and LSC with 1250 superpixels approaches. Thus, six sets of annotations were obtained.

Now, basing on these six annotation sets it will be tested, in terms of e.g. Dice's coefficient, whether LSC1250 yields better agreement than the Manual approach. For this purpose, for each annotated image Dice's coefficient between 1st and 2nd LSC annotation, 1st and 3rd LSC annotation, and 2nd and 3rd LSC annotation were computed. As for each annotated image 3 Dice's coefficient values were obtained, an average value is calculated for LSC method. In a similar fashion an average Dice's coefficient is assigned to each annotated image for a Manual method. Thus, for each annotated image two numbers were considered: an average Dice's coefficient from LSC annotations and an average Dice's coefficient from Manual annotations.

Given these two sets of numbers the next step is testing the significance of the difference between them using an appropriate statistical test which, in this case, is Wilcoxon signed-rank test. The statistical test returns p-value which, if smaller than assumed significance level, indicates that the null hypothesis should be rejected in favor of the alternative hypothesis. The significance level assumed in this study is 5%.

This procedure is described in the following in a more formal way for each hypothesis tested in this study. In particular, four phases of statistical tests are defined below, each referring to a specific objective:

- Phase 1: evaluate whether inter-rater agreement indices are larger when supported by superpixels annotations as compared to the manual approach;

- Phase 2: find which superpixel method achieves the best performances for supporting the annotation procedure;
- Phase 3: evaluate whether the number of superpixels (i.e. lower or higher) affects the annotation process;
- Phase 4: test the intra-rater agreement.

For the inter-rater agreement analysis all cases were annotated with all compared methods so they were treated as paired samples, therefore Wilcoxon signed-rank test was used. In contrast, for intra-rater analysis each physician annotated different set of cases, and for this reason the sets of cases annotated with each method were different. Consequently, Mann–Whitney test was used to analyze the results of intra-reader variability experiment (Rice, 1995).

3.1.1. Phases 1, 2, and 3

A pseudo-code of the proposed pipeline follows, which refers to the assessment of inter-rater agreement and to the identification of the best superpixel method and superpixel number.

Index computation

With $index = \text{DICE, Cohen's Kappa}$

- $Res_A = \{[avg\,(index_{image}\,(ann_i, ann_j)) \quad \text{over} i, j = 1, 2, 3 \quad and \quad i \neq j$] for each annotated 2D $image\}$, with method LSC-lower;
- $Res_B = \{[avg\,(index_{image}\,(ann_i, ann_j)) \quad \text{over} i, j = 1, 2, 3 \quad and \quad i \neq j$] for each annotated 2D $image\}$, with method LSC-higher;
- $Res_C = \{[avg\,(index_{image}\,(ann_i, ann_j)) \quad \text{over} i, j = 1, 2, 3 \quad and \quad i \neq j$] for each annotated 2D $image\}$, with method TPS-lower;
- $Res_D = \{[avg\,(index_{image}\,(ann_i, ann_j)) \quad \text{over} i, j = 1, 2, 3 \quad and \quad i \neq j$] for each annotated 2D $image\}$, with method TPS-higher;
- $Res_E = \{[avg\,(index_{image}\,(ann_i, ann_j)) \quad \text{over} i, j = 1, 2, 3 \quad and \quad i \neq j$] for each annotated 2D $image\}$, with method SLIC-lower;
- $Res_F = \{[avg\,(index_{image}\,(ann_i, ann_j)) \quad \text{over} i, j = 1, 2, 3 \quad and \quad i \neq j$] for each annotated 2D $image\}$, with method SLIC-higher;

- $Res_G = \{[avg\,(index_{image}\,(ann_i, ann_j)) \quad \text{over} i, j = 1, 2, 3 \quad and \quad i \neq j$] for each annotated 2D $image$}, with manual annotation.

With $index =$ Custom Index

- $Res_A = \{index_{image}\,(ann_1, ann_2, ann_3)$ for each annotated 2D $image$}, with method LSC-lower;
- $Res_B = \{index_{image}\,(ann_1, ann_2, ann_3)$ for each annotated 2D $image$}, with method LSC-higher;
- $Res_C = \{index_{image}\,(ann_1, ann_2, ann_3)$ for each annotated 2D $image$}, with method TPS-lower;
- $Res_D = \{index_{image}\,(ann_1, ann_2, ann_3)$ for each annotated 2D $image$}, with method TPS-higher;
- $Res_E = \{index_{image}\,(ann_1, ann_2, ann_3)$ for each annotated 2D $image$}, with method SLIC-higher;
- $Res_F = \{index_{image}\,(ann_1, ann_2, ann_3)$ for each annotated 2D $image$}, with method SLIC-higher;
- $Res_G = \{index_{image}\,(ann_1, ann_2, ann_3)$ for each annotated 2D $image$}, with manual annotation.

Hypothesis testing in Phase 1 - test whether the values of similarity indices from superpixel-supported procedures are larger than from manual approaches

- Test 1: H0: $median\,(Res_A) = median(Res_G)$; H1: $median\,(Res_A) > median(Res_G)$;
- Test 2: H0: $median\,(Res_B) = median(Res_G)$; H1: $median\,(Res_B) > median(Res_G)$;
- Test 3: H0: $median\,(Res_C) = median(Res_G)$; H1: $median\,(Res_C) > median(Res_G)$;
- Test 4: H0: $median\,(Res_D) = median(Res_G)$; H1: $median\,(Res_D) > median(Res_G)$.
- Test 5: H0: $median\,(Res_E) = median(Res_G)$; H1: $median\,(Res_E) > median(Res_G)$.

- Test 6: H0: $median\,(Res_F) = median(Res_G)$; H1: $median\,(Res_F) > median(Res_G)$.

Hypothesis testing in Phase 2 - check which is the best superpixels-based method

- Test 1: H0: $median\,(Res_A) = median(Res_C)$; H1: $median\,(Res_A) \neq median(Res_C)$;
- Test 2: H0: $median\,(Res_A) = median(Res_D)$; H1: $median\,(Res_A) \neq median(Res_D)$;
- Test 3: H0: $median\,(Res_A) = median(Res_E)$; H1: $median\,(Res_A) \neq median(Res_E)$;
- Test 4: H0: $median\,(Res_A) = median(Res_F)$; H1: $median\,(Res_A) \neq median(Res_F)$;
- Test 5: H0: $median\,(Res_B) = median(Res_C)$; H1: $median\,(Res_B) \neq median(Res_C)$;
- Test 6: H0: $median\,(Res_B) = median(Res_D)$; H1: $median\,(Res_B) \neq median(Res_D)$.
- Test 7: H0: $median\,(Res_B) = median(Res_E)$; H1: $median\,(Res_B) \neq median(Res_E)$.
- Test 8: H0: $median\,(Res_B) = median(Res_F)$; H1: $median\,(Res_B) \neq median(Res_F)$.

Hypothesis testing in Phase 3 - check which is the best number of superpixels based on the considered method

- Test 1: H0: $median\,(Res_A) = median\,(Res_B)$; H1: $median\,(Res_A) \neq median\,(Res_B)$;
- Test 2: H0: $median\,(Res_C) = median\,(Res_D)$; H1: $median\,(Res_C) \neq median\,(Res_D)$.
- Test 3: H0: $median\,(Res_E) = median\,(Res_F)$; H1: $median\,(Res_E) \neq median\,(Res_F)$.

3.1.2. Phase 4

A pseudo-code of the proposed pipeline follows, which refers to the assessment of intra-rater agreement.

Index computation

With $index = \text{DICE, Cohen's Kappa}$

- $Res_A = \{ [avg (index_{image} (ann(t_i)_{rater_m}, ann(t_j)_{rater_m}))$ over $i, j = 0, 1, 2$ and $i \neq j$] for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$}, with method LSC-lower;
- $Res_B = \{ [avg (index_{image} (ann(t_i)_{rater_m}, ann(t_j)_{rater_m}))$ over $i, j = 0, 1, 2$ and $i \neq j$] for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$}, with method LSC-higher;
- $Res_C = \{ [avg (index_{image} (ann(t_i)_{rater_m}, ann(t_j)_{rater_m}))$ over $i, j = 0, 1, 2$ and $i \neq j$] for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$}, with method TPS-lower;
- $Res_D = \{ [avg (index_{image} (ann(t_i)_{rater_m}, ann(t_j)_{rater_m}))$ over $i, j = 0, 1, 2$ and $i \neq j$] for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$}, with method TPS-higher;
- $Res_E = \{ [avg (index_{image} (ann(t_i)_{rater_m}, ann(t_j)_{rater_m}))$ over $i, j = 0, 1, 2$ and $i \neq j$] for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$}, with method SLIC-lower;
- $Res_F = \{ [avg (index_{image} (ann(t_i)_{rater_m}, ann(t_j)_{rater_m}))$ over $i, j = 0, 1, 2$ and $i \neq j$] for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$}, with method SLIC-higher;
- $Res_G = \{ [avg (index_{image} (ann(t_i)_{rater_m}, ann(t_j)_{rater_m}))$ over $i, j = 0, 1, 2$ and $i \neq j$] for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$}, with manual annotation.

With $index = \text{Custom Index}$

- $Res_A = \{$ $index_{image} (ann(t_1)_{rater_m}, ann(t_2)_{rater_m}, ann(t_3)_{rater_m})$ for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$ }, with method LSC-lower;

- $Res_B = \{ index_{image} \left(ann(t_1)_{rater_m}, ann(t_2)_{rater_m}, ann(t_3)_{rater_m}\right)$ for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$ }, with method LSC-higher;
- $Res_C = \{ index_{image} \left(ann(t_1)_{rater_m}, ann(t_2)_{rater_m}, ann(t_3)_{rater_m}\right)$ for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$ }, with method TPS-lower;
- $Res_D = \{ index_{image} \left(ann(t_1)_{rater_m}, ann(t_2)_{rater_m}, ann(t_3)_{rater_m}\right)$ for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$ }, with method TPS-higher;
- $Res_E = \{ index_{image} \left(ann(t_1)_{rater_m}, ann(t_2)_{rater_m}, ann(t_3)_{rater_m}\right)$ for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$ }, with method SLIC-lower;
- $Res_F = \{ index_{image} \left(ann(t_1)_{rater_m}, ann(t_2)_{rater_m}, ann(t_3)_{rater_m}\right)$ for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$ }, with method SLIC-higher;
- $Res_G = \{ index_{image} \left(ann(t_1)_{rater_m}, ann(t_2)_{rater_m}, ann(t_3)_{rater_m}\right)$ for 2D $image$ in annotatedImageSet$(rater_m)$ for $m = 1, 2, 3, 4$ }, with manual annotation.

Hypothesis testing Phase 4 - intra-rater agreement

- Test 1: H0: $median\left(Res_A\right) = median(Res_G)$; H1: $median\left(Res_A\right) > median(Res_G)$;
- Test 2: H0: $median\left(Res_B\right) = median(Res_G)$; H1: $median\left(Res_B\right) > median(Res_G)$;
- Test 3: H0: $median\left(Res_C\right) = median(Res_G)$; H1: $median\left(Res_C\right) > median(Res_G)$;
- Test 4: H0: $median\left(Res_D\right) = median(Res_G)$; H1: $median\left(Res_D\right) > median(Res_G)$.
- Test 5: H0: $median\left(Res_E\right) = median(Res_G)$; H1: $median\left(Res_E\right) > median(Res_G)$;

- Test 6: H0: $median\,(Res_F) = median(Res_G)$; H1: $median\,(Res_F) > median(Res_G)$.

3.2. Annotation time

Main problem with acquisition of annotations of medical images is the amount of time spend on their creation. That kind of work requires high expertise in the domain and usually annotations have to be repeated by several physicians. These factors result in difficulties in obtaining the amount of annotated images that would be desired, for example, in machine learning tasks. We believe that any method allowing to speed up this process should be applied. Therefore, the time spent on annotating images is important factor during comparison of different methods. For that purpose time spent on each case during the annotation process was measured by physicians and its average values were compared (method-wise) proving that use of superpixels results in significant speedup of the procedure.

3.3. Manual corrections amount

To compare different superpixeling methods the area of manual corrections made by physicians was measured. During the annotation process physicians applied corrections (addition or subtraction) to the area marked with superpixels to make sure that the borders or annotations are aligned with the borders of lesions. The better the superpixel boundaries match the borders of lesions the less correction is needed, therefore the ratio of the area of those manual corrections to the area marked with superpixels indicates the usability of the superpixeling algorithm for the purpose of annotating.

3.4. Utility for deep learning models training

To check whether improvement of inter-rater agreement in annotations may indeed improve performance of deep learning models based on such annotations, we trained segmentation models based on both manual and superpixel-based annotations. As a metric quality metric of models' performance Dice coefficients were calculated. Only superpixels generated by LSC method were evaluated as this method lead to the highest improvement of inter-rater agreement. Statistical significance of the results was verified with Wilcoxon signed-rank test. To train the segmentation models nnUNet framework (Isensee et al., 2021) was used.

The training and testing data were prepared in the following way. For sacroiliac joints 7 3D cases were randomly selected as the training cases from the total of 9 3D cases available for this study, while the remaining two cases were used for testing. To generate the training data, all lesion annotations (both manual and superpixel-based) were extracted from the annotation images and split into 2D axial slices of the lesions. Ten pixel wide padding was added around each 2D section of each lesion annotation and then corresponding rectangular region was cut from the MR images of the lesion. Patches of MR images together with corresponding patches of manual annotation images were used for training a manual annotation based UNet segmentation model. The training set for this task consisted of 921 pairs of MR patches and corresponding patches of annotation images. Analogously, a training set was prepared for training a superpixel annotation based UNet segmentation model. This training set for this task consisted of 801 pairs of MR and annotation patches. To prepare testing data for both segmentation models (a manual annotation based and a superpixel annotation based) a majority voting annotation was first prepared for each 3D test case. The majority voting was prepared based on all six annotations available for each 3D test case (three manual annotations and three superpixel annotation): whenever a pixel was marked as belonging to a lesion in at least four annotations, it was marked as belonging to a lesion in the majority voting annotation. Given the majority voting annotations for the testing 3D cases, the testing pairs of patches of MR and annotation images were prepared as in the case of training data. The testing set consisted of 36 pairs of MR patches and corresponding patches of majority voting annotations.

In the case of knee dataset 4 3D cases out of 6 3D cases available were randomly selected for training, while the remaining 2 3D cases were used for testing. The training and testing sets were prepared analogously to the case of sacroiliac joints except the step of patches extraction. Because lesions in knee images are relatively big compared to the image size, we used whole 2D axial sections of MR and annotation images in training and testing instead of cutting from these images patches containing lesion. The training set consisted of 93 pairs of MR and annotation images for both the task of training manual annotation based segmentation model and the task of training superpixel annotation based segmentation model. The testing set consisted of 16 pairs of coronal slices of MR and majority voting annotations.

Since the improvement in inter-reader agreement for abdominal images was limited, we have not trained segmentation models for this task.

4. Results

The experiments have shown that application of superpixels in annotation process not only allows to improve inter- and intra-rater reliability but also significantly reduces time spend on annotations. For all tasks use of LSC-lower method has resulted in significantly better reliability, as assessed by the metrics of agreement, than manual annotations. Also all tested methods substantially decreased the annotation time making the annotation procedure almost 2 times faster in some cases. Moreover it was proven that using superpixel-based annotations for training deep learning models leads to improvement of their predictions quality.

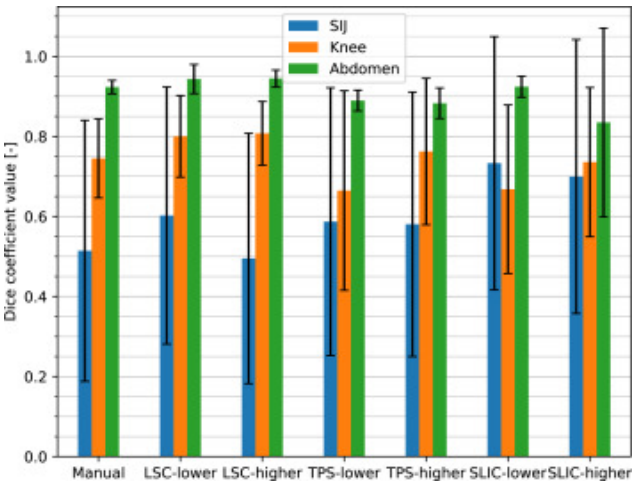
4.1. Inter-rater agreement

The comparison of Dice coefficients for inter-rater agreement is presented in Fig. 4. The values of Cohen's Kappa and Custom Index show the same relations and lead to analogical conclusions. The plots of these metrics can be found in Appendix A. The p-values of statistical tests can be found in Appendix B. As the statistical analysis demonstrates, use of LSC-lower method leads to improvement of inter-rater agreement for all tasks and all indices. Additionally application of TPS-higher and SLIC-higher methods allowed to increase reliability for SIJ and Knee datasets. Use of LSC-higher method increases reliability for knee data while TPS-lower and SLIC-lower methods improved agreement for SIJ data (TPS-lower did not provide improvement in Custom Index). The biggest improvement for SIJ data is obtained with SLIC-lower method - Dice coefficient increases from 0.51 to 0.73. Use of the LSC-higher method resulted in best performance for knee data - Dice coefficient increases from 0.75 to 0.81 - and abdomen data - Dice coefficient increases from 0.92 to 0.94.

The LSC-lower can be considered to be the most universal method as it offered improvement of reliability in all tasks. However use of SLIC method (with both - higher and lower number of superpixels) has resulted in higher improvement of reliability in SIJ task, moreover the difference with TPS-higher method for Knee and SIJ was not statistically significant. Also the difference between methods LSC-

higher and TPS-higher for knee task is not statistically relevant. For SIJ task methods TPS-lower and TPS-higher are statistically better than LSC-higher in terms of Dice coefficient and Cohen's Kappa, however the difference is not significant in terms of Custom Index. Moreover SLIC-lower and SLIC-higher are better than all other methods in that task. On the other hand using LSC-lower and LSC-higher leads to better agreement than TPS-lower and both variants of SLIC method in terms of all indices for knee data. For abdomen data only use of LSC methods allowed to obtain increase of reliability, but it is worth to notice that manual annotations had much higher reliability (Dice coefficient was equal to 0.92) than in other tasks.

Also the preferred amount of superpixels is not unequivocal. For LSC method lower number is better for SIJ task but the number of superpixels is statistically insignificant for knee and abdomen tasks. In contrary, for TPS and SLIC higher number of superpixels is better for knee data and is statistically insignificant for SIJ and abdomen tasks.



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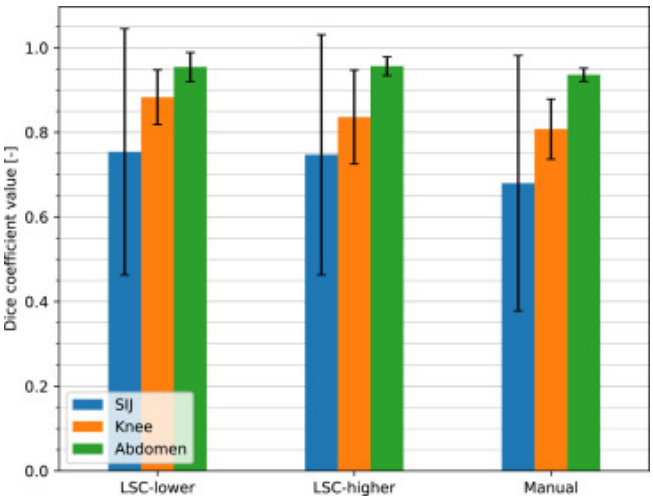
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Fig. 4. Comparison of the mean median value of Dice coefficient for inter-rater agreement (error bars visualize the standard deviation).

4.2. Intra-rater agreement

As it can be seen in Fig. 5 both tested methods offer improvement of intra-rater agreement for all tasks (again as the results are similar for Cohen's Kappa and Custom Index, plots of these indices were moved to Appendix A). The Dice coefficient raised from 0.68 to 0.75 for both LSC-lower and LSC-higher for SIJ data, from 0.81 to 0.88 (0.84) for LSC-lower (LSC-higher) and from 0.94 to 0.95 (0.96) for LSC-lower

(LSC-higher). All the differences are statistically significant (see B.4 in Appendix B).



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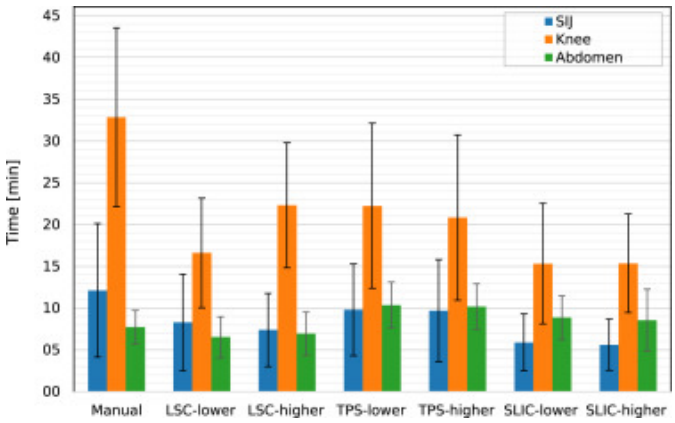
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Fig. 5. Comparison of the mean median value of Dice coefficient for intra-rater agreement (error bars visualize the standard deviation).

4.3. Annotation time

Average time spent on annotation is visualized in Fig. 6. Annotations with use of superpixeling methods were generally more time-effective as they resulted in lower time required to annotate the images in most cases (TPS and SLIC methods were slower for Abdomen task). For SIJ and knee datasets the annotations with SLIC method were the fastest, but also those made basing on LSC and TPS algorithms were still much faster than manual ones. On the other hand LSC method was the only method faster than manual for abdomen task. It seems that generally higher number of superpixels allows to get better increase of annotation speed. LSC-higher was faster than LSC-lower in SIJ task, but slower in knee and abdomen task, while SLIC-higher was faster than SLIC-lower in SIJ and abdomen task, but marginally slower (only 3 s of difference) in knee task. TPS-higher was faster than TPS-lower in all tasks. Nevertheless use of superpixels in annotation process allowed to achieve over 2 times acceleration for knee (15:18 for SLIC-lower method comparing to 32:50 for manual) and SIJ data (5:36 for SLIC-higher vs 12:07 for manual). In abdomen task speedup factor of 1.2 was achieved (6:31 for LSC-lower vs. 7:42 for manual). Relatively high standard deviation which may be noticed is the result of differences between cases - some of them required much more time than others. However, as all

cases was annotated exactly the same number of times, the mean value is still a good metric for comparison.



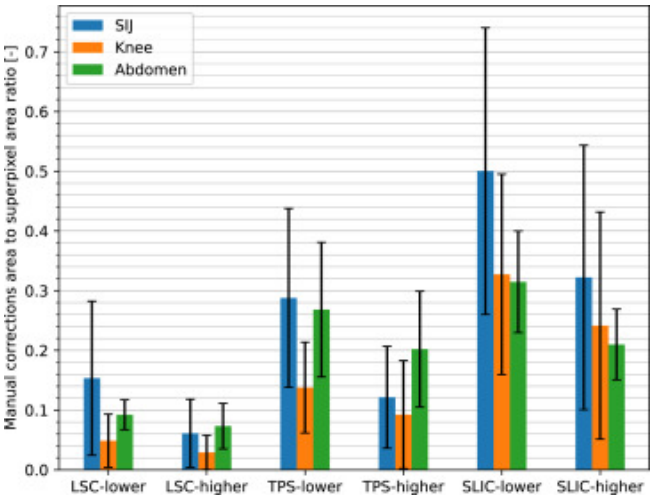
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Fig. 6. Comparison of the average time needed to annotate single case with different methods (error bars visualize the standard deviation).

4.4. Manual corrections amount

The average ratio of manual corrections is shown in Fig. 7. In this matter, the advantage of the LSC method over the others is accentuated. In SIJ annotation task usage LSC-higher method offers the best results - 0.06 compared to 0.12 for TPS-higher and 0.32 for SLIC-higher. For Knee data again LSC-higher method was the best - 0.03 compared to 0.09 for TPS-higher and 0.24 for SLIC-higher. Similarly in abdomen task LSC-higher had the best performance - 0.07 compared to 0.20 and 0.21 for TPS-higher and SLIC-higher methods. In all cases usage of version with lower amount of superpixels resulted in higher ratio of manual corrections. It is also worth to notice that despite higher amount of corrections needed when using SLIC methods the annotations process was still faster in SIJ and Knee task (see par. 4.3), probably due to that, thanks to 3D segmentation, changes on several slices are marked simultaneously. Generally the performance was good but the differences between methods were significant. Moreover a high standard deviation indicates that the alignment of the borders of superpixels with the lesion boundaries is not equally good for every case.



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Fig. 7. Comparison of the average value of manual corrections to superpixel area ratio (error bars visualize the standard deviation).

4.5. Utility for deep learning models training

In the sacroiliac joints task the mean (standard deviation) of Dice coefficient for test data was equal to 0.39 (0.34) for segmentation model based on manual annotations and to 0.54 (0.28) for segmentation model based on superpixel annotations. These results were compared using Wilcoxon test and p-value of 0.005 was obtained (test statistics equal to 114.0), indicating strong statistical evidence for the superiority of the superpixel-based model. Additionally, the lower standard deviation for the superpixel-based model indicates that its performance was more consistent.

In the knee task the mean (standard deviation) of Dice coefficient for test data was equal to 0.69 (0.12) for segmentation model based on manual annotations and to 0.72 (0.08) for segmentation model based on superpixel annotations. The Wilcoxon test's p-value of 0.083 (test statistics equal to 34.0) is a mild evidence of a statistically significant improvement obtained by use of superpixel-based model. Similarly it also allowed to lower the standard deviation of the results, suggesting a more consistent performance.

5. Discussion

The accuracy of statistical learning-based segmentation algorithms is naturally constrained by uncertainties arising from both the data acquisition process (aleatoric uncertainty) and the restricted size of

the training dataset (epistemic uncertainty). Expanding the training dataset and ensuring it closely reflects the data used during inference can help alleviate epistemic uncertainty. However, aleatoric uncertainty, often stemming from factors like noise in image-related signals, remains a persistent challenge and can only be partially reduced.

5.1. Ground truth-related uncertainty

In the case of statistical learning-based segmentation algorithms an important source of aleatoric uncertainty is uncertainty related to ground truth, usually manual segmentations, used at the training time. Consequently, reducing this source of uncertainty is a means of improving the accuracy of segmentation models. Of course, this uncertainty cannot be reduced to zero because it represents different opinions of different raters about same object to be segmented. In fact, the uncertain annotations encompass a huge informative content, as they could represent fuzzy opinions about whether the pathology is just started developing or its effect is mild etc.

Inter- and intra-reader variability in medical data processing is a critical consideration in quantitative analysis and decision-making (Gwet, 2014, Ressman et al., 2019, Mansur et al., 2022). In the case of image segmentation inter-reader variability refers to the differences in segmentation outcomes when performed by different human readers, while intra-reader variability refers to the variability observed when the same reader performs multiple segmentations on the same image. These variabilities can arise due to subjective factors such as differences in experience, visual perception, and interpretation of imaging features. They can also be influenced by objective factors, including the complexity of the image, the quality of the data, and the segmentation methodology employed. Understanding and quantifying inter- and intra-reader variability are essential for assessing the reliability and reproducibility of segmentation results, as well as for optimizing algorithms and protocols to reduce inconsistencies and improve the accuracy and robustness of medical image analysis.

5.2. Dealing with annotations variability

Despite annotations variability is a recognized issue in the literature, the tendency to deal with it is to efficiently combine different annotations using some consensus procedures (Audelan et al., 2022). The annotations variability has been also studied in the context of

deep learning. For example, in Wei et al. (2021) the authors use multiple annotations to train an expertise-aware inferring module, aimed at producing segmentations which mimic the expertise of radiologists with different levels of expertise. In contrast, in Lemay et al. (2022) it is shown that deep segmentation models trained on multiple annotations of the same structure estimate better the segmentation uncertainty than segmentation models trained on fused annotations. The findings of Lemay et al. (2022) support our claim that decreasing annotation uncertainty will lead to better quality predictions. While multiple manual annotations are frequently used to train segmentation models, these manual segmentations are most frequently fused before using them in model training. Moreover, the analysis of intra-reader variability is made frequently to compare discrepancy between automated and ground-truth segmentations against the discrepancies between multiple manual annotations (Rzecki et al., 2021).

The key idea behind the proposed approach is that superpixels, like all oversegmentation algorithms, are expected to gather image pixels or voxels into regions which are homogeneous in some sense. Thus, when preparing annotations, the task of the rater is greatly reduced: instead of manually grouping pixels into annotated regions one has to group superpixels correcting manually such grouping afterwards if necessary. This, in turn, leads to decreased annotation time and increases confidence (in terms of intra- and inter-reader variability) about the boundaries of the objects to be annotated, as demonstrated by the results of our experiments. These lower variability annotations can be used in at least two ways: they can be used to produce more accurate fused ground truth annotations or can be used just as the training data, decreasing variance of the segmentation model predictions.

5.3. Choice of superpixeling method

The main goal of the study was presenting a new approach to reduce inter- and intra-rater variability of annotations of medical images based on superpixel segmentation. A few promising methods of division into superpixels was tested, however wider analysis would be needed to find the best performing method in particular tasks. In the inter-rater agreement experiments, only LSC-lower was consistently better than manual annotation. It is worth noting that the LSC method does not promote superpixel regularity to the same extent as the other methods, which may have allowed for a more accurate

matching of superpixel boundaries to the tissues being labeled. Lower manual corrections to superpixel area ratio for LSC methods seems to be supporting this hypothesis. At the same time, the number of superpixels should not be too high; even if a higher number of superpixels provides better boundary matching, a smaller area of individual superpixels worsens the final result.

In the presented analysis we focuses on application of classical superpixel algorithms. Nevertheless in the recent years some deep learning based algorithms has been proposed (Geusen et al., 2022, Zhu et al., 2021, Yu et al., 2021, Tu et al., 2018). Despite the remarkable success of deep learning methods in various image analysis tasks, the practical deployment of deep learning-based superpixel algorithms for medical image segmentation presents challenges due to their inherent complexity. Deep learning-based algorithms often require adjusting a significant number of hyperparameters which makes them more fragile to domain shift and more difficult to deploy in a superpixel-assisted manual annotation tool. Moreover, the open source codes and pretrained models for the deep learning superpixel algorithms are not always available which makes them difficult in application due to their strong dependence on implementation details. Moreover, no benchmark has yet been established that would reliably compare algorithms of this type, which makes it difficult to choose the appropriate one.

Due to these practical issues relatively older methods were preferred in this study, although deeper analysis of the recent deep learning methods is needed for future work. For the preliminary analysis (Geusen et al., 2022) algorithm was applied to sacroiliac joints dataset (with default values of the parameters), but the performance of the algorithm was quite unpredictable. For some slices superpixels were generated correctly and they could be used in our approach, however for other slices (from the same 3d image) we observed the domination of concentric superpixels which made annotations almost impossible. Moreover, for some slices it was not possible to generate correct superpixel segmentation (there were numerous small and disjoint clusters and a big “background” cluster, no cluster related to object boundaries present in the image). This demonstrated the necessity of tuning the algorithm to a specific dataset before it could be applied to the proposed annotation method.

5.4. Ground truth for segmentation task

The analysis showed that the use of superpixel-based annotations to teach deep segmentation models leads to improvements in their performance. It should be noted, however, that building ground truth for such an assessment is itself a troublesome task. What should be considered to be a gold standard when different approaches to obtain that gold standard are evaluated? In our study we decided to build the ground truth data using a majority voting between both manual and superpixel based annotations. It could be questioned if the final ground truth tends to be closer to the superpixel-based segmentations (which would naturally increase the Dice scores of the model trained with superpixel-based segmentations) or to manual segmentations (increasing the Dice scores of the model trained with manual segmentations).

To mitigate this type of risk, the final ground truths (obtained by majority voting) were compared with 3 superpixel-based and 3 manual segmentations and averaged over each image for each dataset. The resulting Dice scores are not significantly different for superpixel-based and manual segmentation based methods (for sacroiliac joints dataset mean Dice score between manual annotations and majority voting ground truth is 0.50 + /−0.11, mean Dice score between superpixel based annotations and majority voting ground truth is 0.57 + /−0.24, p-value of the Wilcoxon test is 0.176; for knee dataset mean Dice score between manual annotations and majority voting ground truth is 0.78 + /−0.08, mean Dice score between superpixel based annotations and majority voting ground truth is 0.81 + /−0.07, p-value of the Wilcoxon test is 0.105; due to marginal differences between manual and superpixel base annotations for the abdominal dataset the analysis was not performed) which is an argument for validity of the proposed procedure of ground truth generation.

5.5. Superpixels in deep learning

Superpixel-based processing has been considered in deep learning also in other contexts. In studies Fourcade et al., 2020, Song et al., 2014, Chen et al., 2023 and Zhang et al. (2023) deep models are devised that include additional paths which mimic superpixel oversegmentation. The predictions of the “superpixel” paths are fused with predictions of pixel-based paths aiming at improving the segmentation quality. In studies Wei et al. (2022) and Cai et al. (2021) superpixels were used to prepare segmentation masks in a similar manner as in our study but without manual corrections.

Neither of the aforementioned studies has however demonstrated that application of superpixels for ground truth data preparation decreased inter- and intra-reader variability, which is important for the segmentation task, as it is a very basic fact in statistics that for any statistical model the variance of the independent variables influences the variance of the dependent variables. It is a direct consequence of the fact that the deep models are just very high dimensional nonlinear statistical models for which the reduction of prediction variance can be achieved by the means of decreasing variance of inputs (see par. 4.5). Therefore, the investigation of annotation methods that reduce the ground truth variance is beneficial for any task using such data. Note also that as for any statistical models the information about the variance of inputs is necessary to estimate the output variance, we are focusing on annotation improvement but not on annotation fusion (in which case information about input variance is lost).

5.6. Future research and challenges

There are a few research topics which emerge from the proposed approach to annotating. Firstly, further research would be beneficial to determine to what extent the increase in intra- and inter-reader reliability achieved by the proposed approach improves the accuracy of segmentation models. To answer this question, the segmentation problem should be formulated within a probabilistic bayesian framework with multiple annotations of the same image used at the training time: for one training session purely manual segmentations should be used as reference data and for the second training session superpixel-based annotations should be used. Then, the variance of predictions can be estimated at the inference time for both training policies and an appropriate statistical testing can be applied to test the influence of annotation method on the segmentation accuracy. This reduction of prediction variance should be likely analyzed in function of superpixeling method, number of raters, annotation time reduction etc. Next, as the reduction of the intra- and inter-reader variance has been demonstrated for three different tasks and two different modalities, one may ask how the methods of the present article generalize to other tasks and modalities - at this stage of research we left this question open.

An alternative solution to the problems listed above could be different from simply studying the influence on the annotation variability. As it was already mentioned, superpixels group pixels based on some homogeneity criteria. However, superpixel algorithms are quite

generic so the notion of homogeneity used in the definition of the loss function of some superpixel algorithm does not need to fit well to every specific problem. It is clear from our results that none of the experimental configurations (superpixel algorithm and number of superpixels) is generally better than all other experimental configurations even if all lead to decreased annotation uncertainty. This can be however seen as a challenge. Indeed, starting from a generic superpixel-based labeling, annotation uncertainty was already reduced to some extent. Based on the amount of manual corrections to generic superpixel-based annotations one may think about designing specific superpixels with a specific notion of homogeneity fit to a specific task at hand and further reducing annotation uncertainty.

Furthermore, the study compared the agreement of annotations performed by physicians manually and using the proposed method. However, it is worth noting that there is some other tools that can be used to support annotation process like MONAI Label (Diaz-Pinto et al., 2023) or ITK-SNAP (Yushkevich et al., 2016) which could potentially lead to increased reliability. A comprehensive analysis of the effectiveness of available solutions and their comparison with the proposed method would be beneficial and could be the subject of further research.

CRediT authorship contribution statement

Daniel Gut: Conceptualization, Methodology, Project administration, Software, Visualization, Writing – original draft, Writing – review & editing. **Marco Trombini:** Conceptualization, Methodology, Software, Writing – original draft. **Iwona Kucybała:** Data curation, Writing – original draft, Validation. **Kamil Krupa:** Data curation, Validation. **Miłosz Rozynek:** Data curation, Validation. **Silvana Dellepiane:** Conceptualization, Supervision, Writing – original draft, Methodology. **Zbisław Tabor:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Wadim Wojciechowski:** Conceptualization, Data curation, Supervision, Validation.

Declaration of competing interest

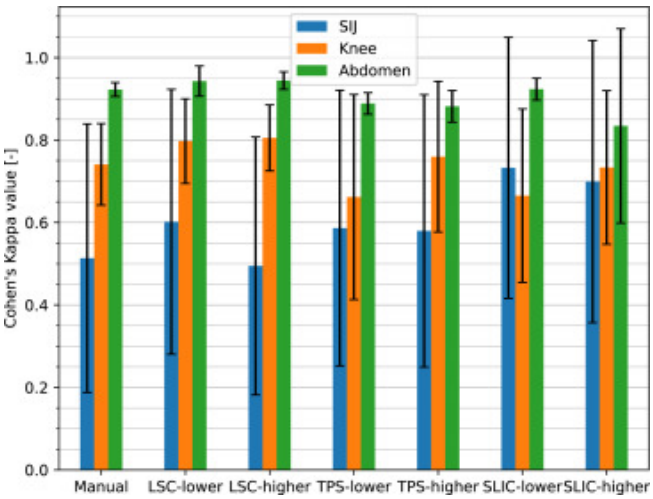
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Additional metrics

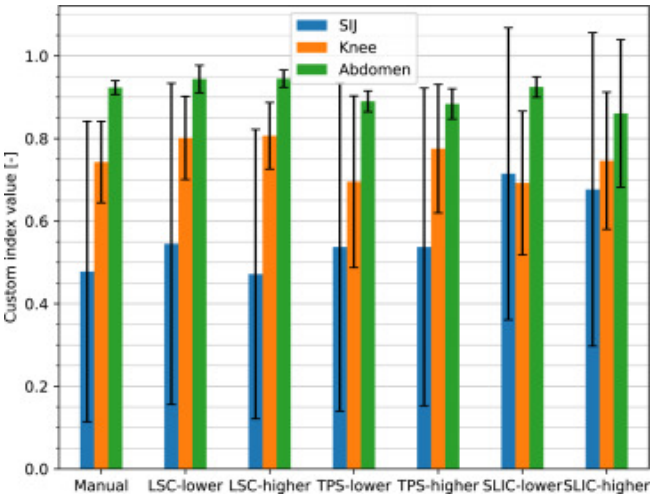
See Fig. A.8, Fig. A.9, Fig. A.10, Fig. A.11.



Download: Download high-res image (268KB)

Download: Download full-size image

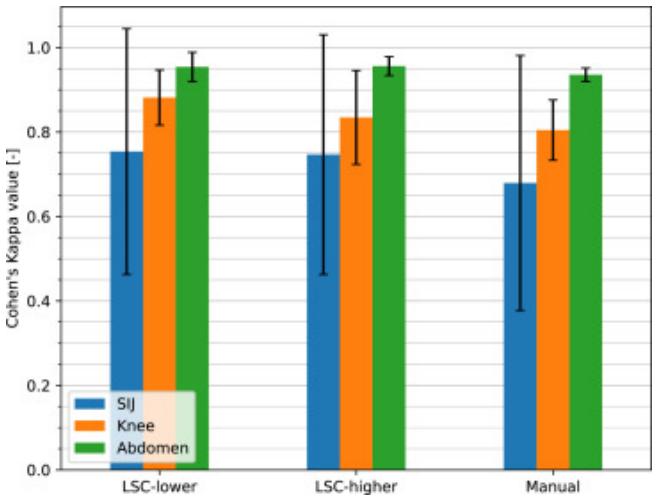
Fig. A.8. Comparison of the mean median value of Cohen's Kappa for inter-rater agreement (error bars visualize the standard deviation).



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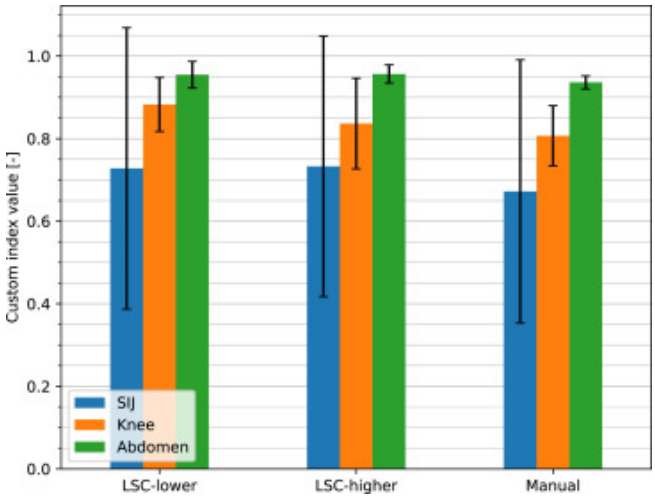
Fig. A.9. Comparison of the mean median value of Custom Index for inter-rater agreement (error bars visualize the standard deviation).



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Fig. A.10. Comparison of the mean median value of Cohen's Kappa for intra-rater agreement (error bars visualize the standard deviation).



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Fig. A.11. Comparison of the mean median value of Custom Index for intra-rater agreement (error bars visualize the standard deviation).

Appendix B. Statistical tests results

See Table B.1, Table B.2, Table B.3, Table B.4.

Table B.1. P-values of Wilcoxon's test comparing median values of metrics obtained with superpixel-based annotations with those obtained with manual annotation performed for inter-rater agreement (H_0: $method = manual$, H_1: $method > manual$).

		LSC-lower	LSC-higher	TPS-lower	TPS-higher	S
Dice	**SIJ**	<0.01	0.66	<0.01	<0.01	<
	Knee	<0.01	<0.01	0.92	<0.01	0
	Abdomen	<0.01	<0.01	1.00	1.00	0
Cohen's Kappa	**SIJ**	<0.01	0.66	<0.01	<0.01	<
	Knee	<0.01	<0.01	0.92	<0.01	0
	Abdomen	<0.01	<0.01	1.00	1.00	0
Custom index	**SIJ**	<0.01	0.45	0.27	0.01	<
	Knee	<0.01	<0.01	0.86	<0.01	0
	Abdomen	<0.01	<0.01	1.00	1.00	0

Table B.2. P-values of Wilcoxon's test comparing median values of metrics obtained with different superpixeling methods.

			TPS-lower	TPS-higher	SLIC-lower	S
Dice	SIJ	**LSC-lower**	0.29	0.36	<0.01	<
		LSC-higher	<0.01	<0.01	<0.01	<
	Knee	**LSC-lower**	<0.01	0.22	<0.01	0
		LSC-higher	<0.01	0.12	<0.01	<
	Abdomen	**LSC-lower**	<0.01	<0.01	<0.01	<
		LSC-higher	<0.01	<0.01	<0.01	<
Cohen's Kappa	SIJ	**LSC-lower**	0.29	0.36	<0.01	<
		LSC-higher	<0.01	<0.01	<0.01	<
	Knee	**LSC-lower**	<0.01	0.22	<0.01	0
		LSC-higher	<0.01	0.12	<0.01	<
	Abdomen	**LSC-lower**	<0.01	<0.01	<0.01	<
		LSC-higher	<0.01	<0.01	<0.01	<
Custom index	SIJ	**LSC-lower**	0.03	0.66	<0.01	<
		LSC-higher	0.21	0.11	<0.01	<
	Knee	**LSC-lower**	<0.01	0.28	<0.01	0

			TPS-lower	TPS-higher	SLIC-lower	S
		LSC-higher	<0.01	0.23	<0.01	<
	Abdomen	**LSC-lower**	<0.01	<0.01	<0.01	<
		LSC-higher	<0.01	<0.01	<0.01	<

			SLIC-lower	SLIC-higher
Dice	SIJ	**TPS-lower**	<0.01	<0.01
		TPS-higher	<0.01	<0.01
	Knee	**TPS-lower**	1.00	0.02
		TPS-higher	<0.01	0.28
	Abdomen	**TPS-lower**	<0.01	<0.01
		TPS-higher	<0.01	<0.01
Cohen's Kappa	SIJ	**TPS-lower**	<0.01	<0.01
		TPS-higher	<0.01	<0.01
	Knee	**TPS-lower**	1.00	0.02
		TPS-higher	<0.01	0.29
	Abdomen	**TPS-lower**	<0.01	<0.01
		TPS-higher	<0.01	<0.01
Custom index	SIJ	**TPS-lower**	<0.01	<0.01
		TPS-higher	<0.01	<0.01
	Knee	**TPS-lower**	0.86	0.04
		TPS-higher	<0.01	0.12
	Abdomen	**TPS-lower**	<0.01	<0.01
		TPS-higher	<0.01	<0.01

Table B.3. P-values of Wilcoxon's test comparing median values of metrics obtained with different number of superpixels (H_0: $lower = higher$, H_1: $lower \neq higher$).

		LSC	TPS	SLIC
Dice	**SIJ**	<0.01	0.64	0.17

		LSC	TPS	SLIC
	Knee	0.79	<0.01	<0.01
	Abdomen	0.51	0.35	0.19
Cohen's Kappa	**SIJ**	<0.01	0.64	0.17
	Knee	0.79	<0.01	<0.01
	Abdomen	0.51	0.35	0.19
Custom index	**SIJ**	0.01	0.06	0.37
	Knee	0.99	<0.01	<0.01
	Abdomen	0.56	0.36	0.18

Table B.4. P-values of Mann–Whitney test comparing median values of metrics obtained with superpixel-based annotations with those obtained with manual annotation for intra-rater agreement (H_0: $method = manual$, H_1: $method > manual$).

		LSC-lower	LSC-higher
Dice	**SIJ**	<0.01	0.01
	Knee	<0.01	<0.01
	Abdomen	<0.01	<0.01
Cohen's Kappa	**SIJ**	<0.01	0.01
	Knee	<0.01	<0.01
	Abdomen	<0.01	<0.01
Custom index	**SIJ**	<0.01	0.02
	Knee	<0.01	<0.01
	Abdomen	<0.01	<0.01

Data availability

In order to ensure reproducibility, custom datasets were made publicly available at https://data.mendeley.com/datasets/425fcppvkn/1 ↗. For task 3 (annotation of psoas major muscles on CT abdominal images) data from Cancer Imaging Archive were used:

https://www.cancerimagingarchive.net/analysis-result/head-neck-ct-atlas/ ↗.

References

Achanta et al., 2010 Achanta R., Shaji A., Smith K., Lucchi A., Fua P., Süsstrunk S.
EPFL (2010)
Google Scholar ↗

Achanta et al., 2012 Achanta R., Shaji A., Smith K., Lucchi A., Fua P., Süsstrunk S.
SLIC superpixels compared to state-of-the-art superpixel methods
IEEE Trans. Pattern Anal. Mach. Intell., 34 (11) (2012), pp. 2274-2282
View in Scopus ↗ Google Scholar ↗

Audelan et al., 2022 Audelan B., Hamzaoui D., Montagne S., Renard-Penna R., Delingette H.
Robust Bayesian fusion of continuous segmentation maps
Med. Image Anal., 78 (2022), Article 102398
View PDF View article View in Scopus ↗
Google Scholar ↗

Ayache and Duncan, 2016 Ayache N., Duncan J.
20Th anniversary of the medical image analysis journal (MedIA)
Med. Image Anal., 33 (2016), pp. 1-3
View in Scopus ↗ Google Scholar ↗

Bergh et al., 2012 Bergh M.V.d., Boix X., Roig G., Capitani B.d., Gool L.V.
Seeds: Superpixels extracted via energy-driven sampling
European Conference on Computer Vision, Springer (2012), pp. 13-26
View in Scopus ↗ Google Scholar ↗

Cai et al., 2021 Cai L., Xu X., Liew J., Sheng Foo C.
Revisiting superpixels for active learning in semantic segmentation with realistic annotation costs
2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, IEEE Computer Society, Los Alamitos, CA, USA (2021), pp. 10983-10992
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Cao et al., 2016 Cao J., Chen Z., Wang B.

Deep convolutional networks with superpixel segmentation for hyperspectral image classification
2016 IEEE International Geoscience and Remote Sensing Symposium, IGARSS, IEEE (2016), pp. 3310-3313
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Chang et al., 2009 Chang H.-H., Zhuang A.H., Valentino D.J., Chu W.-C.
Performance measure characterization for evaluating neuroimage segmentation algorithms
Neuroimage, 47 (1) (2009), pp. 122-135
View PDF View article View in Scopus ↗
Google Scholar ↗

Chen et al., 2018 Chen W., Li W., Dong X., Pei J.
A review of biological image analysis
Curr. Bioinform., 13 (4) (2018), pp. 337-343
Google Scholar ↗

Chen et al., 2017 Chen J., Li Z., Huang B.
Linear spectral clustering superpixel
IEEE Trans. Image Process., 26 (7) (2017), pp. 3317-3330
View in Scopus ↗ Google Scholar ↗

Chen et al., 2023 Chen Y., Li D., Zhang X., Liu P., Meng F., Jin J., Shen Y.
A devised thyroid segmentation with multi-stage modification based on super-pixel U-net under insufficient data
Ultrasound Med. Biol. (2023)
Google Scholar ↗

Dellepiane, 1991 Dellepiane S.
Image segmentation: errors, sensitivity, and uncertainty
Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society Volume 13: 1991, IEEE (1991), pp. 253-254
Crossref ↗ Google Scholar ↗

Diaz-Pinto et al., 2023 Diaz-Pinto A., Alle S., Vishwesh N., Tang Y., Ihsani A., Asad M., Pérez-García F., Mehta P., Li W., Flores M., Roth H.R., Vercauteren T., Xu D., Dogra P., Ourselin S., Feng A., Cardoso M.J.
MONAI label: A framework for AI-assisted interactive labeling of 3D medical images
(2023)
arXiv:2203.12362 ↗

Google Scholar ↗

Fan et al., 2017 Fan J., Chen T., Lu S.
Superpixel guided deep-sparse-representation learning for hyperspectral image classification
IEEE Trans. Circuits Syst. Video Technol., 28 (11) (2017), pp. 3163-3173
Google Scholar ↗

Fourcade et al., 2020 Fourcade, C., Ferrer, L., Santini, G., Moreau, N., Rousseau, C., Lacombe, M., Guillerminet, C., Colombié, M., Campone, M., Mateus, D., Rubeaux, M., 2020. Combining Superpixels and Deep Learning Approaches to Segment Active Organs in Metastatic Breast Cancer PET Images. In: EMBC - Engineering in Medecine and Biology Conference. Montreal, Canada.
Google Scholar ↗

Geusen et al., 2022 Geusen J., Bredell G., Zhou T., Konukoglu E.
Unsupervised superpixel generation using edge-sparse embedding
(2022)
arXiv:2211.15474 ↗
Google Scholar ↗

Grossberg et al., 2017 Grossberg A., Mohamed A., Elhalawani H., Bennett W., Smith K., Nolan T., Chamchod S., Kantor M., Browne T., Hutcheson K., Gunn G., Garden A., Frank S., Rosenthal D., Freymann J., Fuller C.
Data from head and neck cancer CT atlas
(2017), 10.7937/K9/TCIA.2017.umz8dv6s ↗
Google Scholar ↗

Gwet, 2014 Gwet K.L.
Handbook of Inter-Rater Reliability the Definitive Guide to Measuring the Extent of Agreement Among Raters
Advanced Analytics, LLC (2014)
Google Scholar ↗

Isensee et al., 2021 Isensee F., Jaeger P.F., Kohl S.A.A., Petersen J., Maier-Hein K.H.
Nnu-net: a self-configuring method for deep learning-based biomedical image segmentation
Nature Methods, 18 (2) (2021), pp. 203-211
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Kawahara and Hamarneh, 2018 Kawahara J., Hamarneh G.

Fully convolutional neural networks to detect clinical dermoscopic features

IEEE J. Biomed. Health Inform., 23 (2) (2018), pp. 578-585

Google Scholar ↗

Kohli et al., 2017 Kohli M.D., Summers R.M., Geis J.R.

Medical image data and datasets in the era of machine learning—whitepaper from the 2016 C-MIMI meeting dataset session

J. Digit. Imaging, 30 (4) (2017), pp. 392-399

Crossref ↗ View in Scopus ↗ Google Scholar ↗

Lee et al., 2017 Lee G., Tai Y.-W., Kim J.

ELD-net: An efficient deep learning architecture for accurate saliency detection

IEEE Trans. Pattern Anal. Mach. Intell., 40 (7) (2017), pp. 1599-1610

View in Scopus ↗ Google Scholar ↗

Lemay et al., 2022 Lemay A., Gros C., Cohen-Adad J.

Label fusion and training methods for reliable representation of inter-rater uncertainty

J. Mach. Learn. Biomed. Imaging, 31 (2022), pp. 1-29

Google Scholar ↗

Levinshtein et al., 2009 Levinshtein A., Stere A., Kutulakos K.N., Fleet D.J., Dickinson S.J., Siddiqi K.

Turbopixels: Fast superpixels using geometric flows

IEEE Trans. Pattern Anal. Mach. Intell., 31 (12) (2009), pp. 2290-2297

View in Scopus ↗ Google Scholar ↗

Li et al., 2020 Li H., Wei D., Cao S., Ma K., Wang L., Zheng Y.

Superpixel-guided label softening for medical image segmentation

International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer (2020), pp. 227-237

View in Scopus ↗ Google Scholar ↗

Liu et al., 2011 Liu M.-Y., Tuzel O., Ramalingam S., Chellappa R.

Entropy rate superpixel segmentation

CVPR 2011, IEEE (2011), pp. 2097-2104

Crossref ↗ View in Scopus ↗ Google Scholar ↗

Maier-Hein et al., 2018 Maier-Hein L., Eisenmann M., Reinke A., Onogur S., Stankovic M., Scholz P., Arbel T., Bogunovic H., Bradley A.P., Carass A., *et al.*

Why rankings of biomedical image analysis competitions should be interpreted with care
Nat. Commun., 9 (1) (2018), pp. 1-13
Google Scholar ↗

Mansur et al., 2022 Mansur H., Estanislau G., de Noronha M., Marqueti R.d., Fachin-Martins E., Durigan J.L.Q.
Intra- and inter-rater reliability for the measurement of the cross-sectional area of ankle tendons assessed by magnetic resonance imaging
Acta Radiol., 63 (4) (2022), pp. 481-488
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Ouyang et al., 2020 Ouyang C., Biffi C., Chen C., Kart T., Qiu H., Rueckert D.
Self-supervision with superpixels: Training few-shot medical image segmentation without annotation
European Conference on Computer Vision, Springer (2020), pp. 762-780
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Postadjian et al., 2018 Postadjian T., Le Bris A., Mallet C., Sahbi H.
Superpixel partitioning of very high resolution satellite images for large-scale classification perspectives with deep convolutional neural networks
IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium, IEEE (2018), pp. 1328-1331
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Price, 1986 Price K.
Anything you can do, I can do better (no you can't)…
Comput. Vis. Graph. Image Process., 36 (2–3) (1986), pp. 387-391
View PDF View article View in Scopus ↗
Google Scholar ↗

Renard et al., 2020 Renard F., Guedria S., Palma N.D., Vuillerme N.
Variability and reproducibility in deep learning for medical image segmentation
Sci. Rep., 10 (1) (2020), pp. 1-16
Crossref ↗ Google Scholar ↗

Ressman et al., 2019 Ressman J., Grooten W.J.A., Barr E.R.
Visual assessment of movement quality in the single leg squat test: a review and meta-analysis of inter-rater and intrarater reliability

BMJ Open Sport Exerc. Med., 5 (1) (2019)
https://bmjopensem.bmj.com/content/5/1/e000541.full.pdf ↗
Google Scholar ↗

Rice, 1995 Rice J.A.
Mathematical Statistics and Data Analysis
Duxbury Press (1995)
Google Scholar ↗

Rzecki et al., 2021 Rzecki K., Kucybała I., Gut D., Jarosz A., Nabagło T., Tabor Z., Wojciechowski W.
Fully automated algorithm for the detection of bone marrow oedema lesions in patients with axial spondyloarthritis – feasibility study
Biocybern. Biomed. Eng., 41 (2) (2021), pp. 833-853
View PDF View article View in Scopus ↗
Google Scholar ↗

Shi and Malik, 2000 Shi J., Malik J.
Normalized cuts and image segmentation
IEEE Trans. Pattern Anal. Mach. Intell., 22 (8) (2000), pp. 888-905
View in Scopus ↗ Google Scholar ↗

Song et al., 2014 Song Y., Zhang L., Chen S., Ni D., Li B., Zhou Y., Lei B., Wang T.
A deep learning based framework for accurate segmentation of cervical cytoplasm and nuclei
2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (2014), pp. 2903-2906
View in Scopus ↗ Google Scholar ↗

Stutz et al., 2018 Stutz D., Hermans A., Leibe B.
Superpixels: An evaluation of the state-of-the-art
Comput. Vis. Image Underst., 166 (2018), pp. 1-27
View PDF View article View in Scopus ↗
Google Scholar ↗

Suzuki, 2021 Suzuki T.
Implicit integration of superpixel segmentation into fully convolutional networks
(2021)
arXiv preprint arXiv:2103.03435 ↗
Google Scholar ↗

Tu et al., 2018 Tu W.-C., Liu M.-Y., Jampani V., Sun D., Chien S.-Y., Yang M.-H., Kautz J.
Learning superpixels with segmentation-aware affinity loss
2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (2018), pp. 568-576
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Wang et al., 2017 Wang S., Liu L., Duan L., Yu C., Cai G., Gao F., Dong J.
Accurate segmentation of ulva prolifera regions with superpixel and CNNs
2017 International Conference on Security, Pattern Analysis, and Cybernetics, SPAC, IEEE (2017), pp. 433-438
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Wang et al., 2019 Wang J., Zhang C., Gao J., Zhang L.
Brain tumor segmentation from MRI using pre-segmentation based on superpixels and fully convolutional neural networks
2019 IEEE 11th International Conference on Advanced Infocomm Technology, ICAIT, IEEE (2019), pp. 31-35
Google Scholar ↗

Wei et al., 2022 Wei C.C., Mesakh C., Hao C.D., Feipei L., J. L.T., Shen C.Y., Jen C.W.
Deep learning approach based on superpixel segmentation assisted labeling for automatic pressure ulcer diagnosis
PLoS ONE, 17 (2) (2022), pp. 1-19
Crossref ↗ Google Scholar ↗

Wei et al., 2021 Wei J., Shuang Y., Junde W., Kai M., Cheng B., Qi B., Jingjing L., Hanruo L., Li C., Yefeng Z.
Learning calibrated medical image segmentation via multi-rater agreement modeling
2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR (2021), pp. 12336-12346
Google Scholar ↗

Yang et al., 2020 Yang, F., Sun, Q., Jin, H., Zhou, Z., 2020. Superpixel segmentation with fully convolutional networks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 13964–13973.
Google Scholar ↗

Yao et al., 2015 Yao, J., Boben, M., Fidler, S., Urtasun, R., 2015. Real-time coarse-to-fine topologically preserving segmentation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 2947–2955.

Google Scholar ↗

Yu et al., 2021 Yu Y., Yang Y., Liu K.

Edge-aware superpixel segmentation with unsupervised convolutional neural networks

2021 IEEE International Conference on Image Processing, ICIP (2021), pp. 1504-1508

Crossref ↗ View in Scopus ↗ Google Scholar ↗

Yushkevich et al., 2016 Yushkevich P.A., Gao Y., Gerig G.

ITK-SNAP: An interactive tool for semi-automatic segmentation of multi-modality biomedical images

2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC (2016), pp. 3342-3345

Crossref ↗ View in Scopus ↗ Google Scholar ↗

Zhang et al., 2018 Zhang L., Li H., Shen P., Zhu G., Song J., Shah S.A.A., Bennamoun M., Zhang L.

Improving semantic image segmentation with a probabilistic superpixel-based dense conditional random field

IEEE Access, 6 (2018), pp. 15297-15310

Crossref ↗ View in Scopus ↗ Google Scholar ↗

Zhang et al., 2023 Zhang H., Lin M., Yang G., Zhang L.

ESCNet: An end-to-end superpixel-enhanced change detection network for very-high-resolution remote sensing images

IEEE Trans. Neural Netw. Learn. Syst., 34 (1) (2023), pp. 28-42

Google Scholar ↗

Zhu et al., 2021 Zhu L., She Q., Zhang B., Lu Y., Lu Z., Li D., Hu J.

Learning the superpixel in a non-iterative and lifelong manner

2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR (2021), pp. 1225-1234

Crossref ↗ View in Scopus ↗ Google Scholar ↗

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